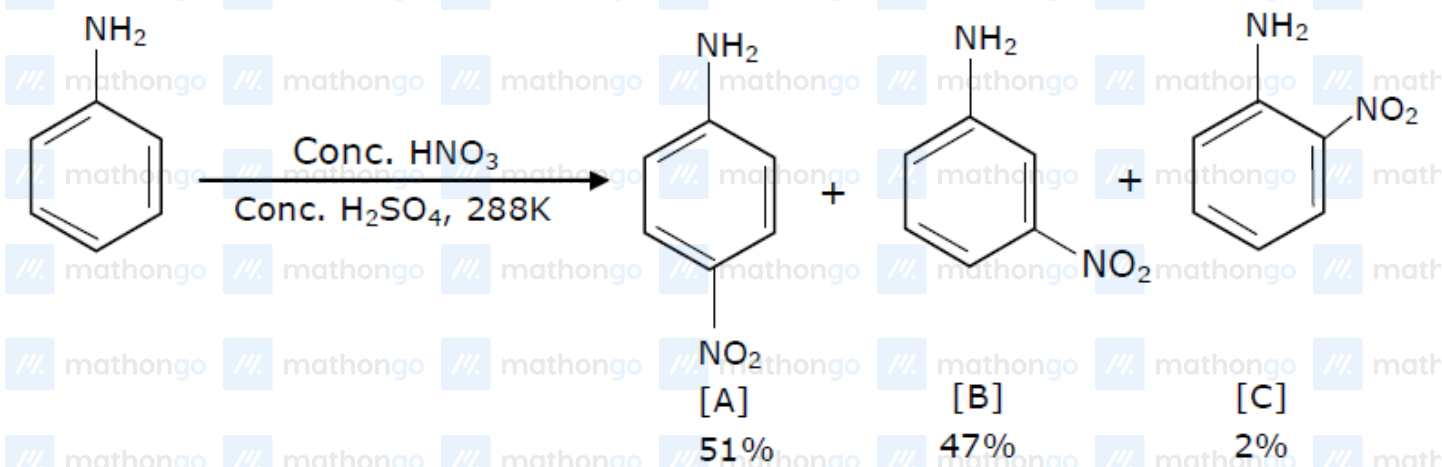


Q1: 24 Feb (Shift 1) - Single Correct

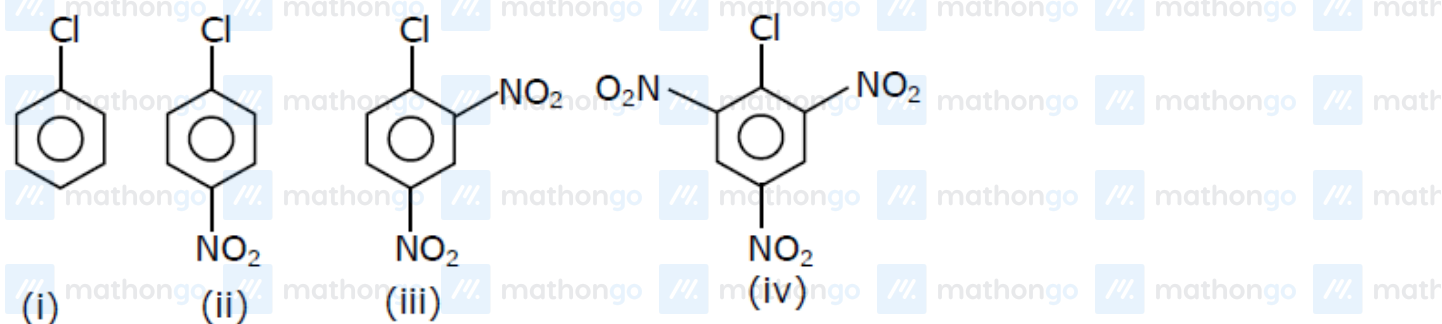
In the following reaction the reason why meta-nitro product also formed is:



- (1) Formation of anilinium ion
- (2) $-\text{NO}_2$ substitution always takes place at meta-position
- (3) low temperature
- (4) $-\text{NH}_2$ group is highly meta-directive

Q2: 24 Feb (Shift 2) - Single Correct

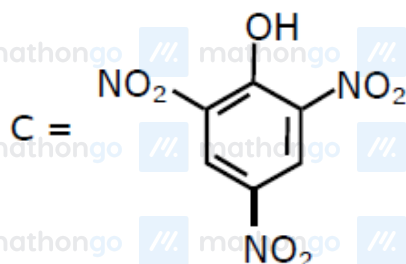
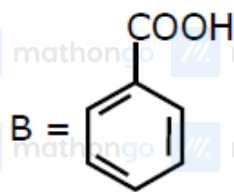
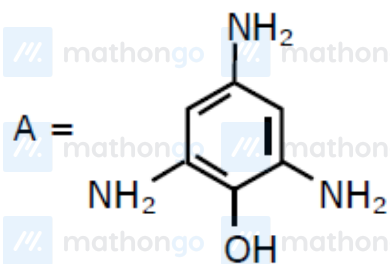
The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is :



- (1) $(iv) < (i) < (iii) < (ii)$
- (2) $(iv) < (i) < (ii) < (iii)$
- (3) $(i) < (ii) < (iii) < (iv)$
- (4) $(iv) < (iii) < (ii) < (i)$

Q3: 25 Feb (Shift 1) - Single Correct

Compound(s) which will liberate carbon dioxide with sodium bicarbonate solution is/are:



(1) B and C only

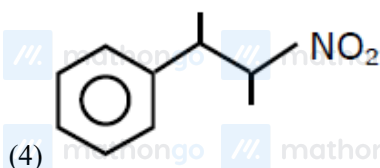
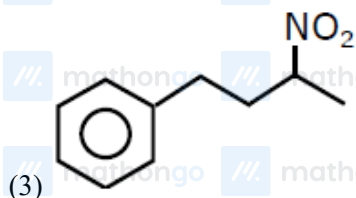
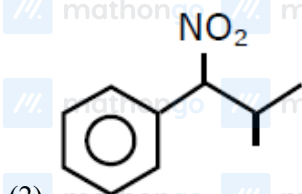
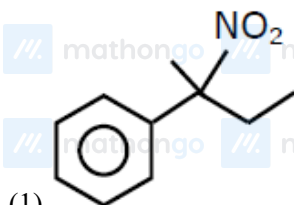
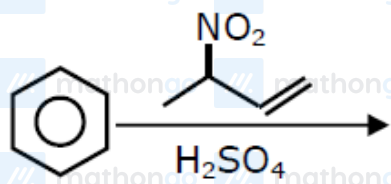
(2) B only

(3) A and B only

(4) C only

Q4: 25 Feb (Shift 2) - Single Correct

The major product of the following reaction is :



Q5: 25 Feb (Shift 2) - Single Correct

The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is :

- (1) $\text{CH}_3\text{COCl}/\text{AlCl}_3, \text{Br}_2/\text{AlBr}_3, \text{HNO}_3/\text{H}_2\text{SO}_4, \text{Zn}/\text{HCl}$
- (2) $\text{CH}_3\text{COCl}/\text{AlCl}_3, \text{Zn} - \text{Hg}/\text{HCl}, \text{Br}_2/\text{AlBr}_3, \text{HNO}_3/\text{H}_2\text{SO}_4$
- (3) $\text{Br}_2/\text{AlBr}_3, \text{CH}_3\text{COCl}/\text{AlCl}_3, \text{HNO}_3/\text{H}_2\text{SO}_4, \text{Zn}/\text{HCl}$
- (4) $\text{HNO}_3/\text{H}_2\text{SO}_4, \text{Br}_2/\text{AlCl}_3, \text{CH}_3\text{COCl}/\text{AlCl}_3, \text{Zn} - \text{Hg}/\text{HCl}$

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Answer Key

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Q1 (1)

Q2 (3)

Q3 (1)

Q4 (4)

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Q5 (2)

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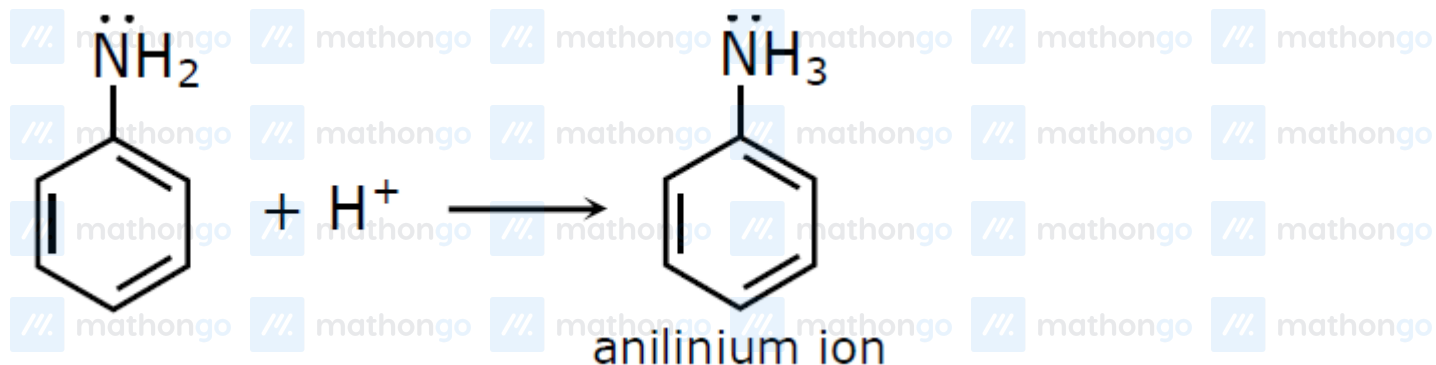
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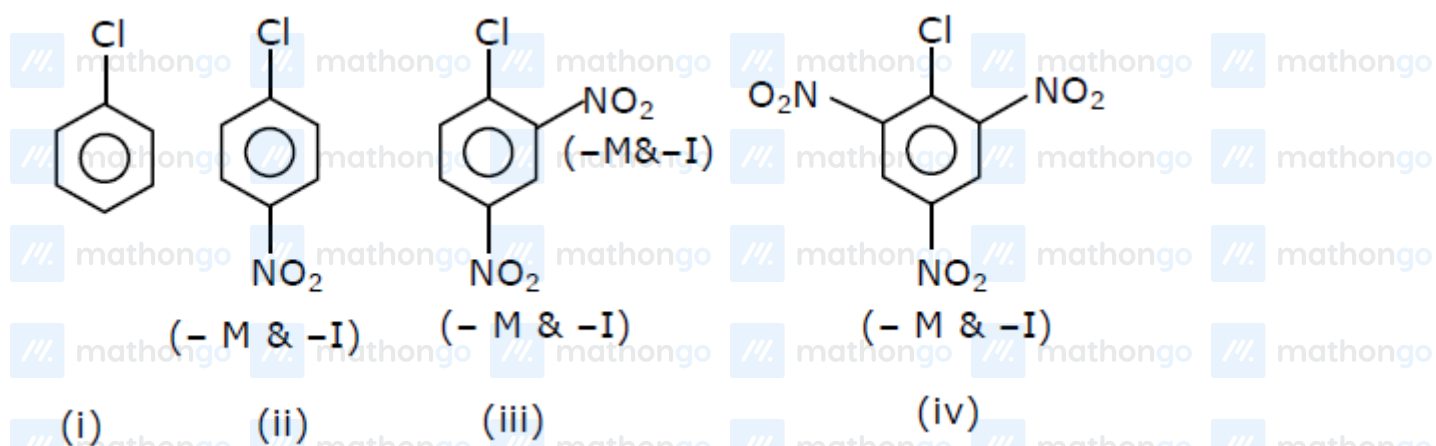
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Q1 (1)



In acidic medium the $-NH_2$ group in aniline converts into anilinium ion which is meta directing.

Q2 (3)



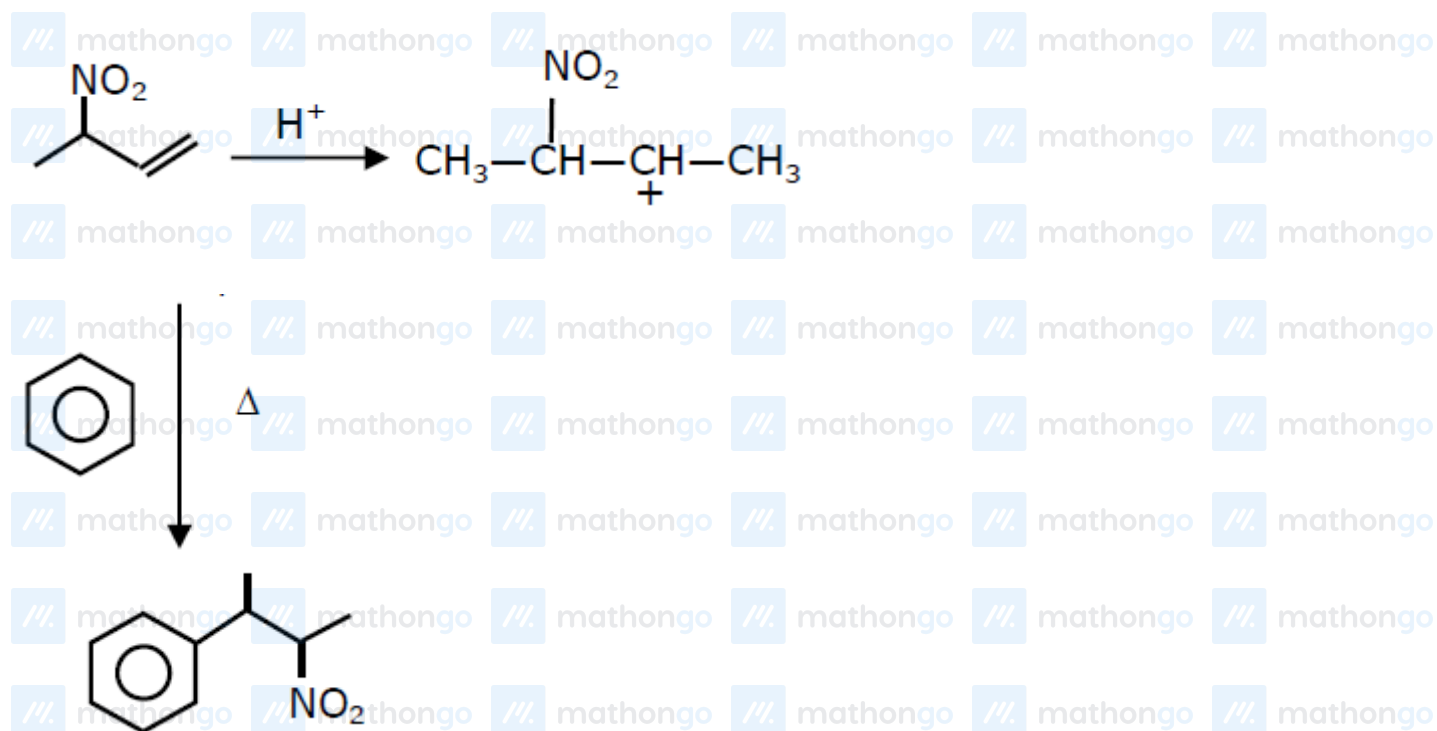
Reactivity $\propto -m$ group present at O/P position.

Q3 (1)

Compounds which are more acidic than H_2CO_3 , gives CO_2 gas on reaction with $NaHCO_3$. Compound B i.e.

Benzoic acid and compound C i.e. picric acid both are more acidic than H_2CO_3 .

Q4 (4)



Q5 (2)

